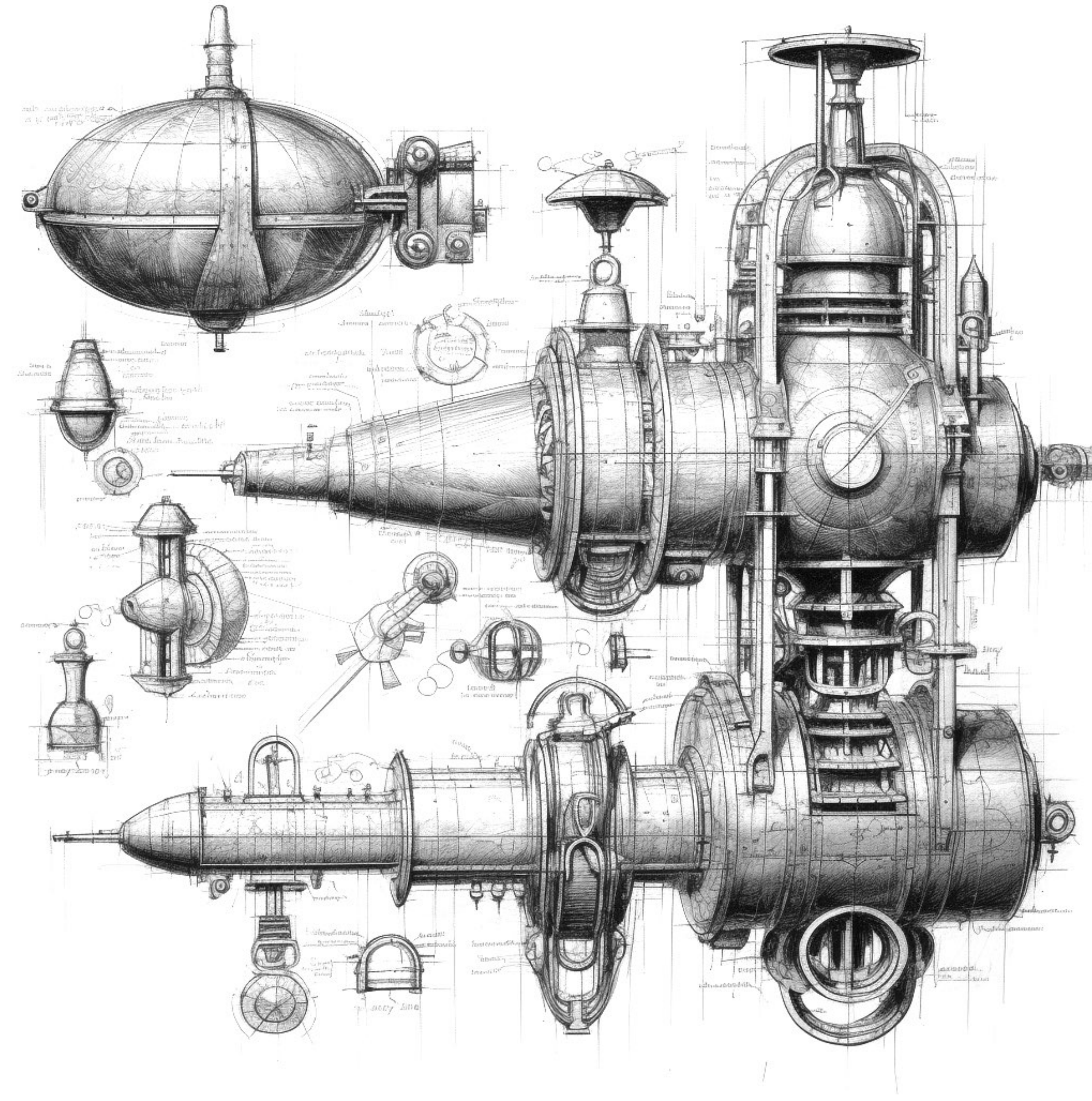




MGT 7316 C: Requirements Engineering

Agenda

June 17, 2023

Business Problems

Requirements Engineering

Key Terms

Techniques

Product Requirements Document

Overview

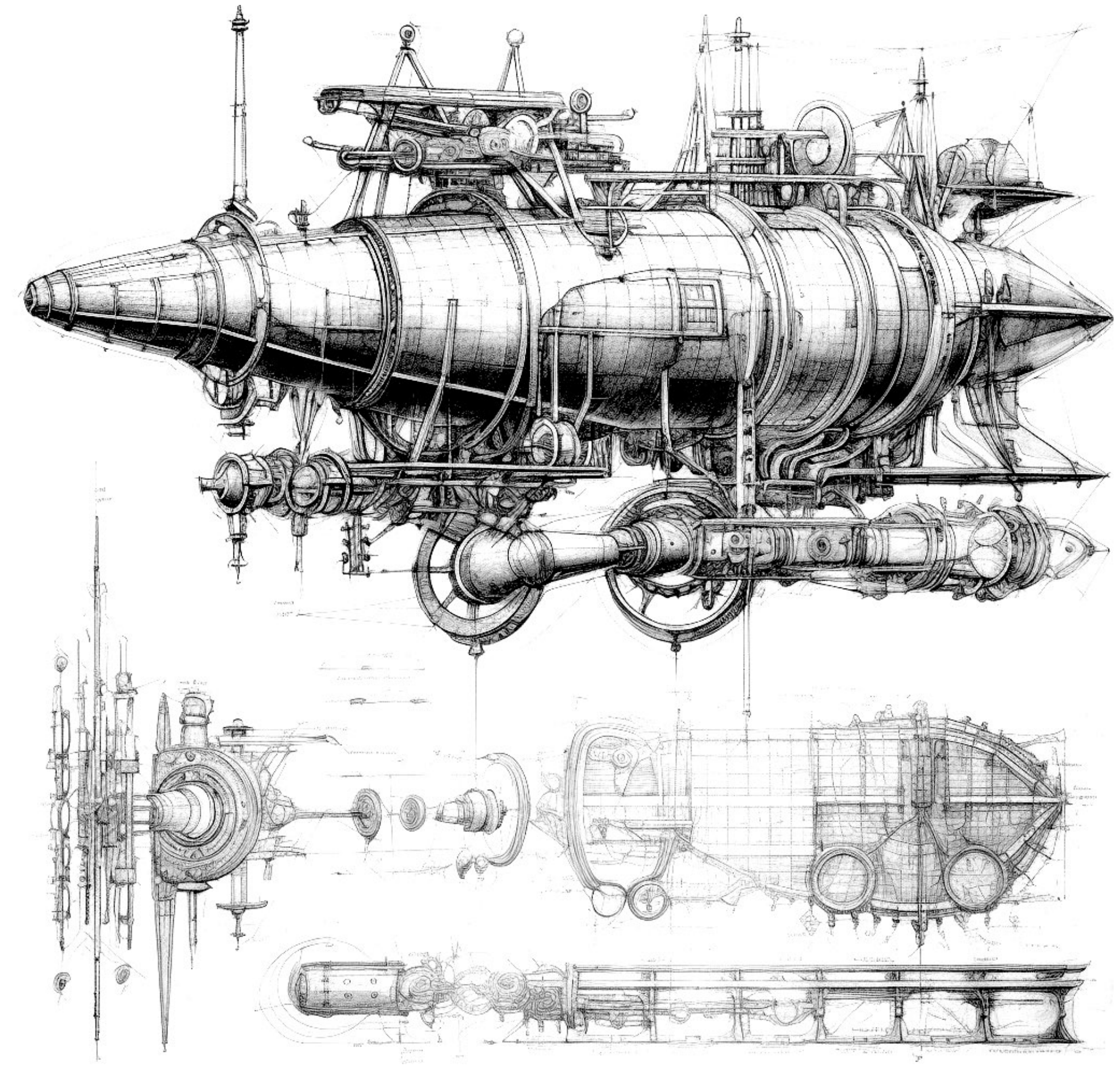
Components

Markdown

Syntax

Replit IDE

Module 1 Tasks



Key Terms



Coding Syntax

requirement - a singular documented physical or functional need that a particular design, product or process aims to satisfy

functional requirements - defines a function where a function is described as a specification of behavior between inputs and outputs

non-functional requirements (nfr) - criteria used to judge the operation of a system, rather than specific behaviors

(software) engineering - design process to design, develop, maintain, test, and evaluate computer software

requirement engineering (re) - process of defining, documenting, and maintaining requirements in the engineering design process

software development life cycle (SDLC) - process for planning, creating, testing, and deploying an information system

Source: <https://en.wikipedia.org/wiki/Requirement>

Source: https://en.wikipedia.org/wiki/Functional_requirement

Source: https://en.wikipedia.org/wiki/Non-functional_requirement

Source: https://en.wikipedia.org/wiki/Software_engineering

Source: https://en.wikipedia.org/wiki/Requirements_engineering

Source: https://en.wikipedia.org/wiki/Systems_development_life_cycle

Business Problem Statements

Problem Components and Solving Techniques

Problem Statement - a concise description of an issue to be addressed or a condition to be improved upon (includes ideal state, reality, consequences, and proposed solution(s))

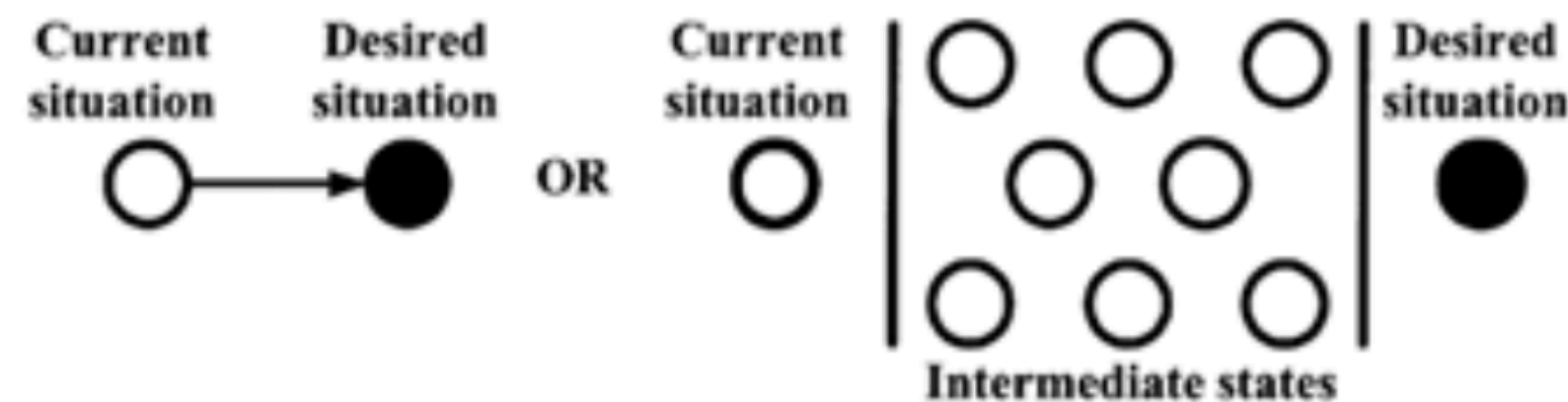


Figure 1: Initial state of Problem solving

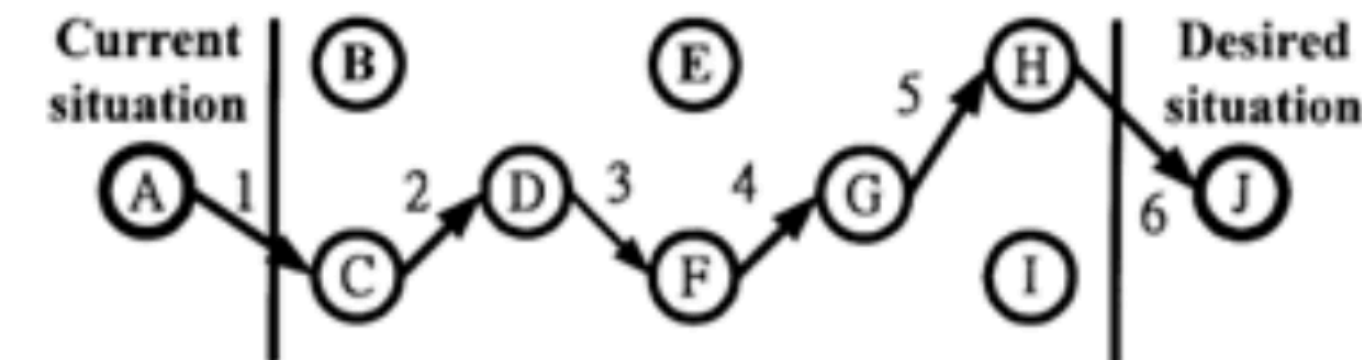


Figure 2: Problem Solving with a chain of intermediate states

Intermediate States - examples include strategic thinking, critical thinking, design thinking, and computational thinking

Strategic Thinking - mental or thinking process applied by an individual in the context of achieving a goal or set of goals

Critical Thinking - analysis of available facts, evidence, observations, and arguments to form a judgment

Design Thinking - an iterative process that teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test

Source: https://en.wikipedia.org/wiki/Problem_statement

Source: [Importance of Problem Statement in Solving Industry Problems, doi:10.4028/www.scientific.net/AMM.421.857](https://doi.org/10.4028/www.scientific.net/AMM.421.857)

Source: https://en.wikipedia.org/wiki/Strategic_thinking

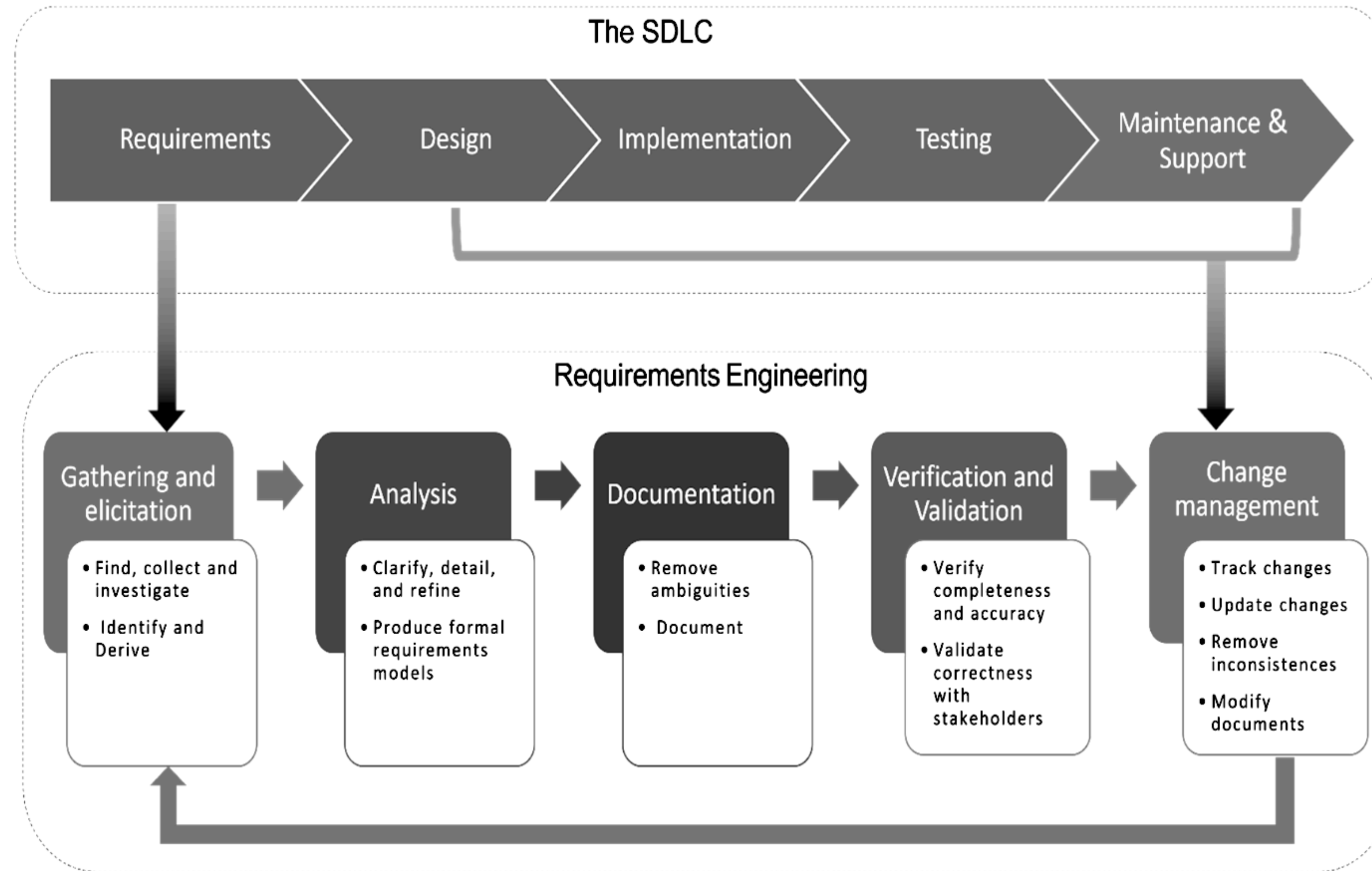
Source: https://en.wikipedia.org/wiki/Critical_thinking

Source: https://en.wikipedia.org/wiki/Design_thinking#As_a_process_for_innovation

Source: <https://www.interaction-design.org/literature/topics/design-thinking>

Key Terms

RE Technique



Source: Kanneh, S., & Anu, V. (2022). Security Requirements Prioritization Techniques: A Survey and Classification Framework. Software, 1(4), 450-472. <https://doi.org/10.3390/software1040019>

Key Terms

Product Requirements Document (PRD) Structure

title - project name

change history - log of what, who, when the document was changed

overview - brief description

acceptance criteria - what are the success metrics (definition of done)

personas - what are the key identities and/or characteristics of the intended audience

release timeline - project timeline and schedule

functional & non-functional requirements - prioritized feature set

user stories - an event-based script that the browser runs in the

design considerations - references to sketches, style guidelines, branding, etc.

other considerations / axioms / assumptions - catch all for any other items or factors in decision-making



Source: <https://productschool.com/blog/product-strategy/product-template-requirements-document-prd>

Key Terms

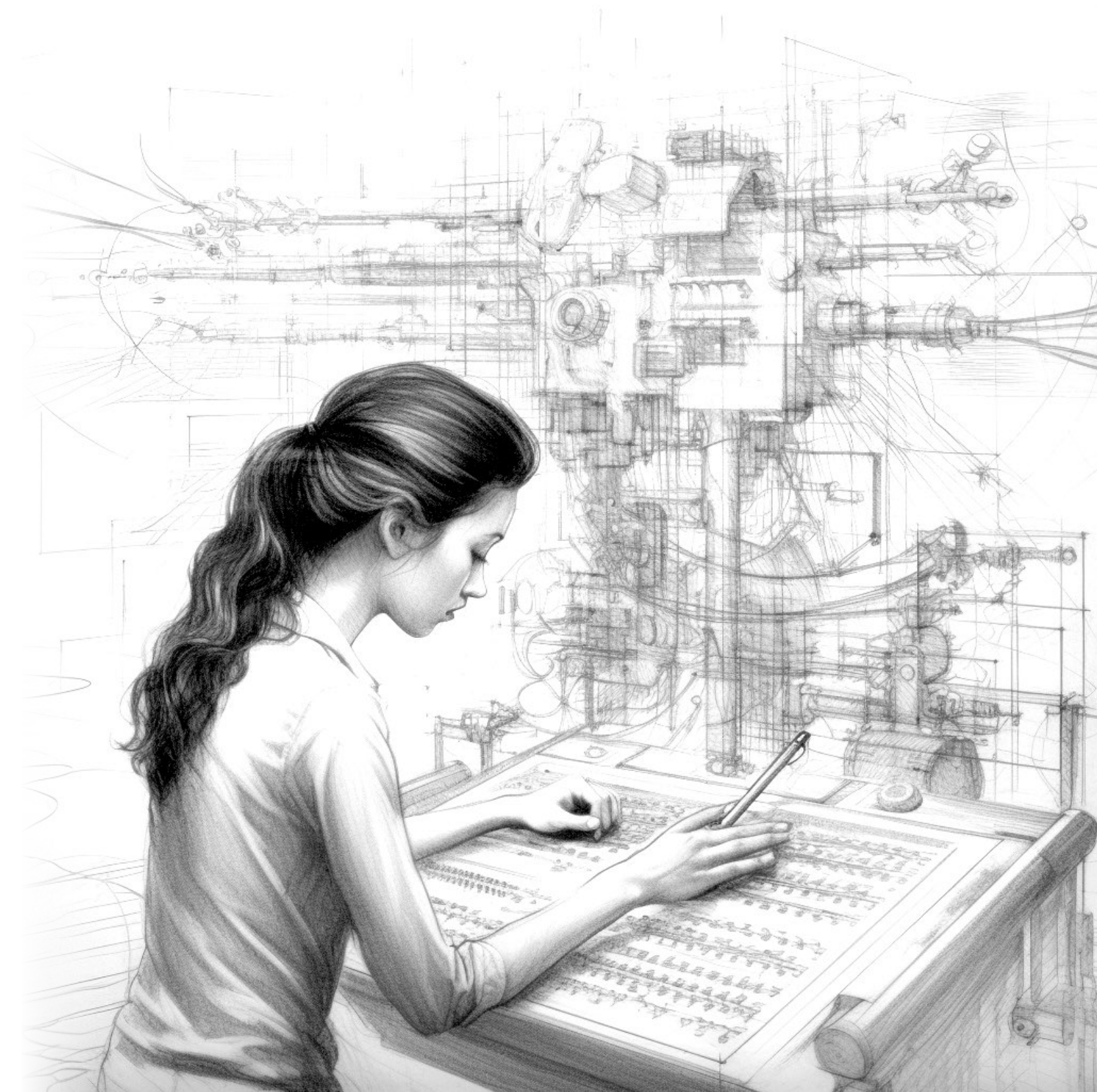
Problem Solving Methodology

1. Create a **problem statement**
2. Use computational thinking as the process which **produces an algorithm/pseudocode** (plain language description of steps) which generates the desired condition or result
3. Convert algorithms/pseudocode (**aka encode**)

In-class exercise: <https://replit.com/@johnmcswain/MGT7316PRD>

Source: <https://en.wikipedia.org/wiki/Pseudocode>

Source: <https://rogerlmartin.com/lets-read/the-design-of-business>



Module 1 Tasks

Readings

[Product Requirements Docs \(Atlassian\)](#)

[Google Technical Writing One](#)

[Markdown Guide](#)

Module 1 Discussion

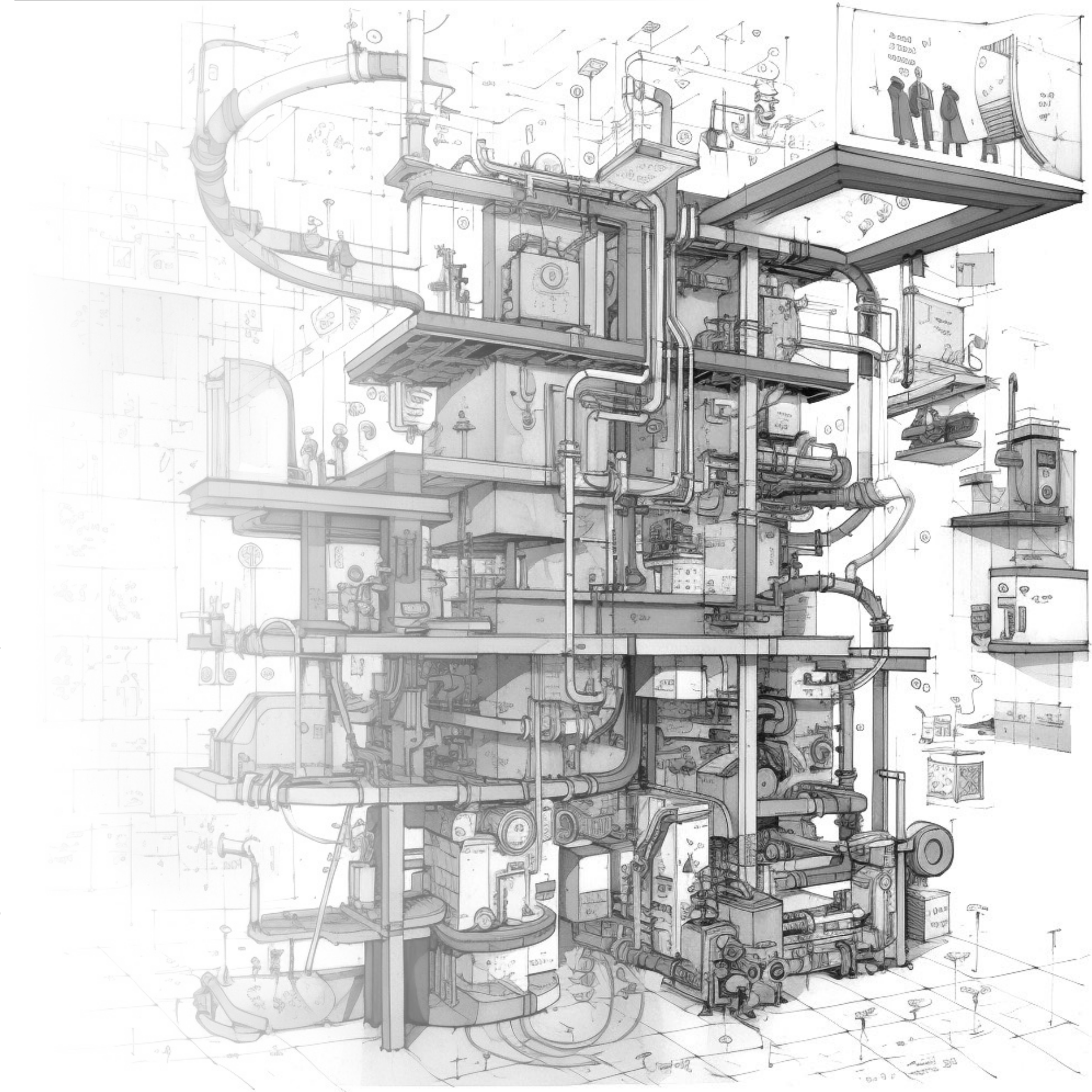
https://canvas.instructure.com/courses/7100862/discussion_topics/18617763

Module 1 Assignment

<https://canvas.instructure.com/courses/7100862/assignments/38364328>

Pre-assessment Quiz

<https://canvas.instructure.com/courses/7100862/assignments/38364328>



Questions?